A P-RING CONDITION CAN BE CHECKED AT MAXIMAL IDEALS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Property P of a ring together with the closure conditions (C) and (D) used in the section. stated in the section, source lines 12800-12952, before the counted statement. Supplies the two rules that let the property be moved along regular maps and factored through localisations.
- Faithful flatness of the completion and the existence of a prime of the completion over a given prime of the localisation. used at proof lines 12961-12970. Produces the prime at which the diagram is drawn.
- Regularity of the maps appearing in the square, in particular that completions of Noetherian local rings have regular formal fibres and that regular maps compose. the author cites Proposition proposition-Noetherian-complete-G-ring and Lemma lemma-regular-composition at proof lines 12975-12985. Supplies the regularity statements that condition (C) consumes.

Source reference. Original lemma 0BIU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14502–14502. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07PS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9807–9807. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07QI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9833–9833. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a Noetherian ring. Assume P satisfies (C) and (D). Then R is a P-ring if and only if the formal fibres of $R_{\mathfrak{m}}$ have P for every maximal ideal $\mathfrak{m}$ of R.*

Proof. Assume the formal fibres of $R_{\mathfrak{m}}$ have P for all maximal ideals $\mathfrak{m}$ of R. Let $\mathfrak{p}$ be a prime of R and choose a maximal ideal $\mathfrak{p} \subset \mathfrak{m}$. Since $R_{\mathfrak{m}} \to R^\wedge_{\mathfrak{m}}$ is faithfully flat we can choose a prime $\mathfrak{p}'$ of $R^\wedge_{\mathfrak{m}}$ lying over $\mathfrak{p}R_{\mathfrak{m}}$. Consider the commutative diagram

$$\begin{array}{ccccc} R^\wedge_{\mathfrak{m}} & \longrightarrow & (R^\wedge_{\mathfrak{m}})_{\mathfrak{p}'} & \longrightarrow & (R^\wedge_{\mathfrak{m}})^\wedge_{\mathfrak{p}'} \\ \uparrow & & \uparrow & & \uparrow \\ R_{\mathfrak{m}} & \longrightarrow & R_{\mathfrak{p}} & \longrightarrow & R^\wedge_{\mathfrak{p}} \end{array}$$

By assumption the fibres of the ring map $R_{\mathfrak{m}} \to R^\wedge_{\mathfrak{m}}$ have P. By Proposition 07PS $(R^\wedge_{\mathfrak{m}})_{\mathfrak{p}'} \to (R^\wedge_{\mathfrak{m}})^\wedge_{\mathfrak{p}'}$ is regular. The localization $R^\wedge_{\mathfrak{m}} \to (R^\wedge_{\mathfrak{m}})_{\mathfrak{p}'}$ is regular. Hence $R^\wedge_{\mathfrak{m}} \to (R^\wedge_{\mathfrak{m}})^\wedge_{\mathfrak{p}'}$ is regular by Lemma 07QI. Hence the fibres of $R_{\mathfrak{m}} \to (R^\wedge_{\mathfrak{m}})^\wedge_{\mathfrak{p}'}$ have P by (C). Since $R_{\mathfrak{m}} \to (R^\wedge_{\mathfrak{m}})^\wedge_{\mathfrak{p}'}$ factors through the localization $R_{\mathfrak{p}}$, also the fibres of $R_{\mathfrak{p}} \to (R^\wedge_{\mathfrak{m}})^\wedge_{\mathfrak{p}'}$ have P. Thus we may apply (D) to see that the fibres of $R_{\mathfrak{p}} \to R^\wedge_{\mathfrak{p}}$ have P. □